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Key Points:

- For the past 25 years, ice cover along Lake Superior's shoreline has been characterized by pronounced interannual variability
- We reproduce these dynamics in two probabilistic models (Cox survival and beta) combining satellite, in situ, and teleconnection data
- Coastal ice conditions along the shoreline of Earth's largest freshwater lake reflect a regime shift that began in the late 1990s

Supporting Information:

Supporting Information may be found in the online version of this article.

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Combining Satellite, Teleconnection, and In Situ Data to Improve Understanding of Multi-Decadal Coastal Ice Cover Dynamics on Earth's Largest Freshwater Lake

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Abstract To differentiate and understand drivers behind coastal ice cover trends and variability, we advance development of a model combining satellite, in situ, and teleconnection data along the shoreline of Earth's largest freshwater lake (Lake Superior). Previous studies suggest a regime shift in Lake Superior's ice cover starting in 1998. Our study includes seven years of new data and subsequent model analysis that provide new insight into characteristics of the post-1998 regime. In addition to providing a valuable extension to the historical ice cover record for this domain, we find the regime shift in coastal ice cover starting in 1998 is characterized by pronounced variability, and not simply a shift in pre-1998 trends. Our findings represent an important stepping stone for future ice and climate modeling not only on Lake Superior but across the entire Great Lakes region and in other global high-latitude coastal regions as well.

Plain Language Summary Ice cover on the Laurentian Great Lakes has become more variable over the past 25 years. To better understand this variability, we re-develop a model from a previous study focused on a portion of the shoreline of Lake Superior. The previous study, which culminated in 2015, suggested a change in both interannual variability and long-term trends in ice cover after 1998. At the time of the previous study, however, the extent to which those changes might continue to propagate into the future was unclear. Here, we extend the historical ice cover record through 2022 while also exploring a broad range of potential explanatory variables in a simulation model. Our analysis indicates that the post-1998 regime is characterized by more pronounced variability than previous studies indicated, with near-record-high years of ice cover followed by years of very little or even no appreciable ice cover. These interannual ice cover dynamics were not evident in the historical record prior to 1998, and their persistence from 1998 through 2022 underscores the importance of not only differentiating regime shifts from trends in climate change studies, but also of the value in correctly reflecting those regime shifts in simulation and forecasting models.

1. Introduction

Understanding changes in ice cover along marine and freshwater coastlines is a fundamental mission of many coastal management planning agencies, including those that oversee and provide guidance for the commercial shipping industry, recreational activities (such as ice fishing and swimming), habitat restoration, and infrastructure design (E. J. Anderson et al., 2018; Comiso et al., 2008). Here, following protocols in previous studies that incrementally update and evaluate long-term data sets and models (J. Wang et al., 2018), we advance findings from a previous study (Ji et al., 2019) that addressed this challenge for a unique coastal region within the Laurentian Great Lakes (hereafter referred to simply as the Great Lakes). The Great Lakes are an ideal geographic domain for a study on climate change and its impacts on coastal hydroclimate processes; a broad range of environmental variables in the region have been shown to be increasingly susceptible to manifestations of climate variability, including, notably, coastal ice cover (Assel, 1991; Gronewold et al., 2013; Howk, 2009; Kistner et al., 2018). The propagation of climate change through different modes of variability in coastal ice cover (including the timing of onset, the timing of breakup, and the fate and transport of surface ice sheets) on the Great Lakes presents a threat to human and ecological health and safety, poses challenges to coastal management planning, and underscores the need for reliable seasonal ice cover forecasts across multiple scales (Assel et al., 2004; Durnford et al., 2018).

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Within this context, our study focuses on improving seasonal ice cover forecasting along the Apostle Islands National Lakeshore (APIS). The APIS is managed by the National Park Service and is located along the southwestern shoreline of Lake Superior (for details, see Krumenaker, 2005). It includes a section of coastline and offshore islands known for their natural beauty and, in winter, for their intricate ice formations (including a network of coastal ice caves). Winter visitors to the APIS can typically only access these caves on foot by crossing ice along the Lake Superior shoreline. Starting in 1998, the areal extent of ice cover in this area has become much more variable (for studies documenting similar dynamics in other areas of the Great Lakes, see Bai et al., 2015; Mason et al., 2016). This variability has, in certain years, limited access to the APIS ice caves while also making it more difficult for APIS management personnel to predict, prior to each winter season, whether ice cover will be adequate for pedestrian access.

Here, we present improvements to a seasonal coastal ice modeling system intended not only to benefit the APIS, but to serve as a foundational tool for ice cover modeling in other freshwater and marine coastal areas as well. The statistical models within this system complement other commonly used models that explicitly simulate lake ice and other physical processes (Grant et al., 2021; Sharma et al., 2020; Woolway & Merchant, 2019; Zdorovennova et al., 2023). They also fit squarely within a broader body of research aimed at improving understanding of how global climate change propagates into variability in physical hydroclimate processes at regional scales (Ghanbari & Bravo, 2008; Lettenmaier et al., 1999; Lofgren & Gronewold, 2013), and how that understanding can improve management decisions, ensure the stability of ecosystem function and aquatic habitat, and protect human health and safety.

2. Methods

Our approach builds on methods introduced in Ji et al. (2019), including the continued development of two statistical regression models; one (a survival or hazard model) designed to simulate the date of ice cover onset in a given year, and a second (a beta model) designed to simulate the probability that there will be any safe ice cover at any point in a given year. In Ji et al. (2019), these models were developed with the somewhat limited goal of exploring the potential explanatory power of teleconnection indices alone on ice cover variability over a data record starting in 1973 and ending in 2015. We note that while the following sections include details of our updated methodology, readers interested in further details of the mathematics behind the Cox and beta models should refer to Ji et al. (2019).

Here, we advance the previous approach in several important ways. First, we include a broader range of potential model explanatory variables, above-and-beyond the teleconnection indices evaluated in Ji et al. (2019). These new variables, described in further detail below, include new teleconnection indices that recent studies suggest are perhaps more (relative to previously considered indices) strongly linked to ice cover dynamics (Fu & Steinschneider, 2019; Lin et al., 2022; Mishra et al., 2011). The new explanatory variables also include regional in situ meteorological data, which were not included in the Ji et al. (2019) study because managers at the APIS were (at that time) specifically interested in the potential for ice cover forecasting tools using climate teleconnection indices alone.

Second, we extend the time period for all model variables, including satellite-based coastal ice cover, through 2022. This step is critical to providing insight into whether or not the supposed ice cover regime that began in 1998 has continued. It is informative to note that other hydroclimate variables across the Great Lakes region (in addition to ice cover) changed dramatically in the late 1990s including, for example, lake evaporation (Assel, 2005) and lake storage (Gronewold & Stow, 2014). The changes in these two variables, however, appear to have reversed starting in 2014, with previously documented abrupt decreases in evaporation and record-setting increases in lake levels from 2014 to 2020 (Gronewold et al., 2015; Matthias et al., 2016). We know of no recent study that has documented the extent to which changes in ice cover duration and onset date have continued across any region of the Great Lakes after the winter of 2015 while also identifying potential physical mechanisms across the entire period of record that would explain both the late 1990s transition and its persistence.

2.1. Data Collection

We collected gridded satellite-based ice cover data across a portion of the Lake Superior shoreline (including the APIS) from the National Oceanic and Atmospheric Administration (NOAA) Great Lakes Environmental Research Laboratory (GLERL) Great Lakes Ice Cover Database, derived from extensions to and reprocessing of the Great Lakes Ice Atlas (Assel, 2005; J. Wang, Assel, et al., 2012; Yang et al., 2020). We extracted data from December 1972 through May 2022, resulting in a new 50-year long continuous regional coastal ice cover record from ice year 1973–2022. Here (and in the NOAA GLERL Great Lakes Ice Cover Database), an "ice year" is defined as a 6-month period from December to the following May. The first ice year of record in the Great Lakes Ice Atlas is 1973, which covers the period from December 1972 through May 1973.

We then extended the time period of record for monthly (August through December prior to each ice year) average teleconnection indices used in Ji et al. (2019) through ice year 2022 [including the Arctic Oscillation (AO), Southern Oscillation Index (SOI), North Atlantic Oscillation (NAO), El Niño-Southern Oscillation (ENSO), Pacific Decadal Oscillation (PDO), and Pacific North American index (PNA)] from the NOAA National Center for Environmental Information (NCEI). This period of record is extended through ice year 2022 such that teleconnection indices from August 2021 to December 2021, for example, would be used to simulate and forecast ice cover dynamics in ice year 2022.

We supplemented this data set with monthly average values for the Atlantic Multi-decadal Oscillation (AMO), the East Pacific/North Pacific pattern (EP-NP), the Outgoing Longwave Radiation (OLR) anomaly index (Liebmann & Smith, 1996), and the Tropical/Northern Hemisphere pattern (TNH) obtained from the National Weather Service's Climate Prediction Center (for EP-NP, TNH, and OLR) and the Physical Sciences Laboratory at NOAA (for AMO). It is informative to note that EP-NP pattern indices were obtained only for the months of August to November (in each year of our record), while TNH was obtained only for the month of December. OLR and AMO were collected for the months of August through December. We also note that the OLR record was incomplete for several ice seasons. In these cases, we in-filled data with corresponding values from the previous calendar year, based on the assumption of a strong OLR seasonal cycle. For example, we find that August to November values of OLR in 2009 were missing, and we in-filled data during that period with corresponding values from August to November of 2008.

In addition to developing a new set of teleconnection indices, we added three regional in situ meteorological variables to our analysis. The new regional meteorological variables include air temperature (T_a), Lake Superior average surface water temperature (T_w), and accumulated winter season severity index (AWSSI). We obtained T_a values from Weather Station USW00014913 at the Duluth (MN) International Airport (one of the long-term continuous stations closest to the APIS) via the NOAA NCEI Global Historical Climatology Network (GHCN) repository (Peterson et al., 1998). We obtained T_w values by combining records from two data sets maintained by NOAA GLERL. The first includes model simulations from 1972 through 1994 from the Large Lake Thermodynamics Model (LLTM) (Croley & Assel, 1994). The second includes data from the Great Lakes Surface Environmental Analysis (or GLSEA, see Schwab et al., 1999, for details). The GLSEA is generally believed to be more accurate than the LLTM, however, it is not available prior to 1994. We compared (results not shown) overlapping periods between the two products (i.e., after 1995) and found that they were nearly identical (for a similar comparison, applied to Lake Michigan, see Gronewold et al., 2015). The AWSSI was obtained from publicly available data records at the Midwest Regional Climate Center (for details, see Boustead et al., 2015). This collective set of updated teleconnection and meteorological data was then used to evaluate potential explanatory variables in our two models (described below) to simulate the timing of ice cover onset and, respectively, the probability of ice cover onset at any point in a given ice year.

2.2. Model Development

We developed multiple variants of two models for simulating and forecasting two respective characteristics of seasonal ice cover. The first was a Cox survival (or hazard) model (Read & Vogel, 2016; Therneau & Grambsch, 2000), designed to simulate the timing (quantified as the number of days after December 1) of ice cover onset in each ice year. It is informative to note that the Ji et al. (2019) study evaluated two survival models, Cox and Weibull, but only the Cox model was recommended for further research because the Weibull model resulted in very wide (relative to the Cox model) uncertainty bounds that were impractical for decision making.

The second model we developed was a beta model (Cribari-Neto & Zeileis, 2010) designed to simulate the probability that safe ice cover conditions would occur at least once, at any point in time, during a given ice season. Following Ji et al. (2019), we defined "safe" ice cover as ice coverage in over 90% of grids in our study area for 10 consecutive days. It is informative to note that the Ji et al. (2019) study recommended that a beta regression model be re-evaluated specifically to explore the potential for fewer explanatory variables; the beta model from that study included far too many variables for what might be considered practical for real-world seasonal forecasting.

We note also that the Ji et al. (2019) study included only 16 years of data following what was believed (as noted previously) to have been a regime shift in the late 1990s (Assel et al., 2000; Rodionov & Assel, 2003; Van Cleave et al., 2014). Here, we develop the Cox and beta models using an expanded historical record that represents a roughly 50% increase in the amount of data following the regime shift. We expect this expanded record to lead to better explanatory power in our models (including a need for fewer variables) and new insight into drivers behind Great Lakes ice cover dynamics.

Prior to model calibration, we performed auto- and cross-correlation analyses on all potential explanatory variables (Gelman & Hill, 2007). If the correlation coefficient between two variables exceeded 0.65, one was excluded from the regression analysis (Weisberg, 2005).

2.2.1. Cox Survival Model

We calibrated the Cox survival model in three ways (i.e., developed three Cox model variants) using the `Surv` and `coxph` functions in the `R` statistical software (R Core Team, 2022) `survival` package (P. Anderson & Gill, 1982; Therneau & Grambsch, 2000). First, we employed stepwise automated regression with the `step` function (Venables & Ripley, 2002), recognizing that this approach (while often used and considered conventional) can lead to different results depending on "stopping rules" and other selection criteria (Bendel & Afifi, 1977).

Second, we calibrated the Cox model using a manual selection procedure in which we systematically removed explanatory variables with a regression coefficient that had a p-value greater than 0.05, until we were left with a model for which all coefficients have p-values less than 0.05 (Gelman & Hill, 2007). We recognize that there are calls within the statistical modeling community for considering more stringent selection criteria (Benjamin et al., 2018), such as using coefficients with p-values less than 0.01. For this study, we proceeded with what we consider to still be conventional practice, and view evaluation of alternate stopping rules and coefficient selection criteria as an area for future research.

Third, and finally, we developed a model in which the explanatory variables (and their corresponding coefficients) were selected first based on findings from previous research (Fu & Steinschneider, 2019; Lin et al., 2022; Mishra et al., 2011). These variables were then refined after determining which, if any, were considered statistically significant. Additional details of the code for this procedure, including `R` console output, are included in Supporting Information S1.

We compared the three Cox models using a combination of concordance, commonly defined as the area under the receiver operator curve (Gönen & Heller, 2005; Hajian-Tilaki, 2013), and a visual comparison between the model simulated "earliest probable ice onset date" and the observed ice onset date in each ice year. Following decision making criteria outlined by APIS personnel, we define the "earliest probable ice onset date" as the date on which, based on output of each Cox model, there is only a 0.1 probability the ice onset would have occurred earlier. Put differently, over the course of our study, we would expect the observed date of ice cover onset to occur *after* our "earliest probable onset date" about 90% of the time.

2.2.2. Beta Probability Model

We then developed and tested three beta probability model variants, also using `R`, following the same basic steps outlined above for the Cox model. First, we conducted an automated stepwise regression procedure on a beta model object constructed using the `gamlss` (generalized additive models for location scale and shape) function (Rigby & Stasinopoulos, 2005; Stasinopoulos et al., 2017). We then conducted a manual stepwise regression procedure using the function `betareg` (Cribari-Neto & Zeileis, 2010) and the model coefficient selection criteria (i.e., using coefficient p-values) outlined above. Third, we developed a model using the same set of explanatory variables used for the third Cox model variant, as identified in Mishra et al. (2011), Fu and Steinschneider (2019), and Lin et al. (2022).

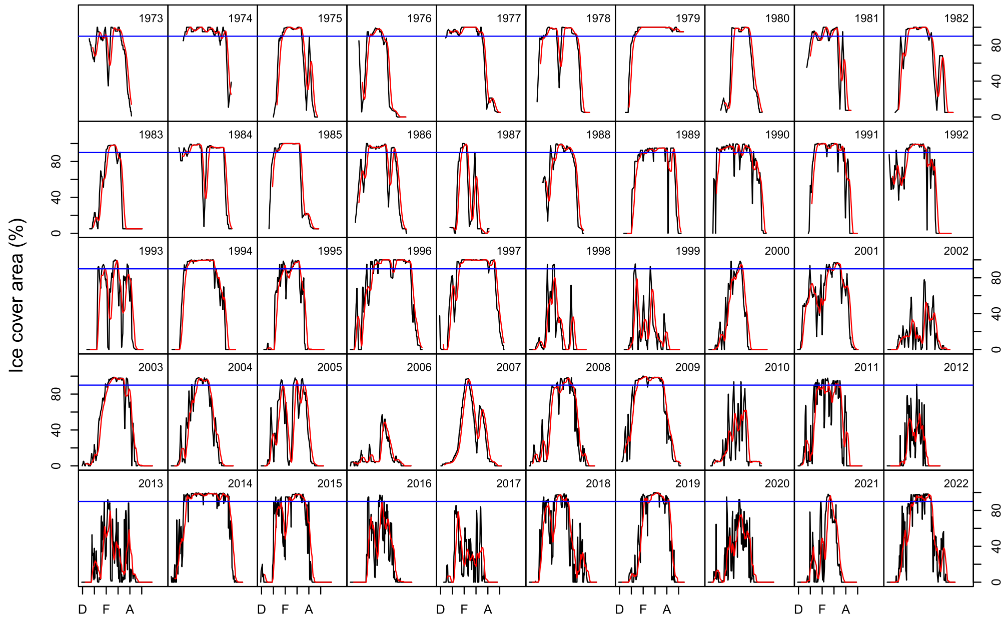

Figure 1. Observed daily ice cover percent areal coverage (black lines) along the southwestern shoreline of Lake Superior (near the APIS), the 10-day rolling average of the observed daily ice cover (red lines), and the pedestrian access safety threshold (90%) for the 10-day rolling average (horizontal blue line) for each year in our study.

2.3. Model Selection and Validation

We selected from among each set of models (i.e., the Cox and beta models) an optimal model that combined a high skill score with a low number of explanatory variables, and we then validated that model by recalibrating it separately for the pre- and post-1998 periods, and by conducting a leave-one-out cross validation (Efron & Tibshirani, 1993) across the model for the entire study period, and within each of the separately calibrated periods. This approach allows for the possibility that two separate models might have greater historical simulation and forecasting skill than a single model over the entire historical period. This approach is also intended to provide additional insight into the potential for any model from our study to be used in real-world seasonal ice cover forecasting applications.

3. Results and Discussion

3.1. Historical Ice Cover

Our new record of daily ice cover (Figure 1 and Figure S1 in Supporting Information S1) along the southwest shoreline of Lake Superior (near the APIS), which extends from 1973 through 2022, indicates that conditions suitable for pedestrian access to the APIS ice caves occurred in every year from 1973 to 1997 and have occurred only 15 times in the 25 years since 1997. Importantly, safe ice cover conditions have occurred only 4 times (2018, 2019, 2021, 2022) between 2016 and 2022; the 7 year period following the most recent study of this area (Ji et al., 2019).

It is worth noting that two of the years with safe ice cover conditions since 1998 (2014 and 2018) coincided with a widespread outbreak of very cold arctic air (i.e., a deformation of the Arctic polar vortex, as described in Butler et al., 2020; Clites et al., 2014; Zhang et al., 2016). This finding suggests that while the post-1998 regime might be characterized, on average, by later ice onset and less ice cover (Mason et al., 2019; L. Wang et al., 2015), it also appears to be punctuated by pronounced variability driven by continental-scale air circulation dynamics (Lee & Butler, 2020; Matthias et al., 2016); an important finding for guiding future model forecasting and management planning in the Great Lakes region.

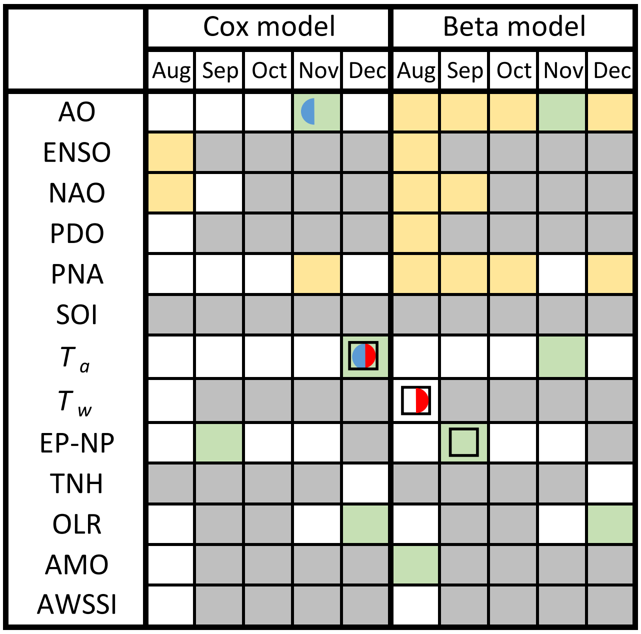

Figure 2. Summary of variables evaluated and retained by manual selection, regression analysis, and expert opinion. Gray cells indicate variables that we removed due to significant cross- or auto-correlation, or which were not available in the original source data archive. Empty white cells indicate variables that we included in the regression analysis but were not found to be statistically significant. Green cells indicate variables noted for their explanatory power in previous ice cover studies. Open black squares indicate variables found to be statistically significant using a manual selection procedure. Blue half-circles indicate variables retained after calibrating each model to only the 1973 to 1997 time period, while red half-circles indicate variables retained after calibrating each model to only the 1998 to 2022 period. Cells with light yellow shading indicate variables included in final models from the Ji et al. (2019) study, for reference.

3.2. Model (Variable and Coefficient) Selection

Our analysis of correlation within and between potential explanatory variables indicates that many are highly cross- or auto-correlated and should be removed from consideration prior to model development (gray cells in Figure 2). ENSO, PDO, T_w, AMO and AWSSI, for example, are all strongly auto-correlated across all 5 months preceding each ice season, and therefore we utilized only the August value of these variables in our analysis. We selected the August monthly value for these variables, in part, to provide as early a lead time as possible for planning decisions in forecasting applications of our model.

The results of our Cox model development and selection procedure indicate that four variables have potential explanatory value for coastal ice cover based on previous ice cover studies (including November AO, December T_a, September EP-NP, and December OLR). Across our entire period of record, only one of these (December T_a) was found to be significant through our manual stepwise selection procedure (Figure 2). When developing the Cox model for the pre-1998 period alone, two of these variables were found to be significant (November AO and December T_a) while only one (December T_a) was significant for the post-1998 period. A closer inspection of the uncertainty in the Cox model coefficients (left column, Figure 3) indicates that interpretation of each coefficient's significance is very sensitive to selected stopping rules, and that it seems reasonable to use all four variables for further model analysis (below).

The results of our beta model development and selection procedure, however, indicate more of a discrepancy between the utility of variables identified from previous studies and those identified through our manual stepwise selection procedure (Figure 2). More specifically, we find that only one of the variables selected in previous studies appears to have utility for our beta model (September EP-NP), and the explanatory power of these variables does not increase when the data set is divided into pre-1998 and post-1998 periods (Figure 2). We do find, however, that August T_w is a significant explanatory variable for models run over the entire historical period, and over the post-1998 period alone. Therefore, our further analysis of the beta model is based on using August T_w and September EP-NP as explanatory variables. We note that one of the selected teleconnection variables, September EP-NP, appears to change modality near the time of the 1997–1998 ice cover regime shift (Figure 4 and Figure S5 in Supporting Information S1).

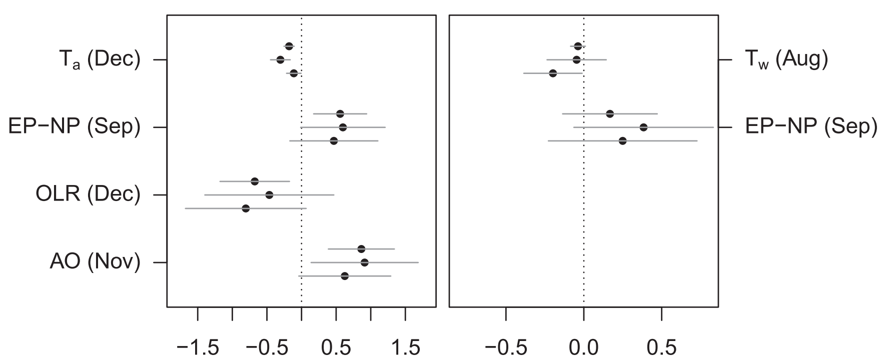

Figure 3. 95% Confidence intervals (gray lines) and median values (black dots) for Cox (left) and beta (right) model coefficients for each selected variable. The three intervals and median values for each coefficient are derived from calibration across the entire data record (top interval and median value), the pre-1998 period (middle), and the post-1998 period (bottom).

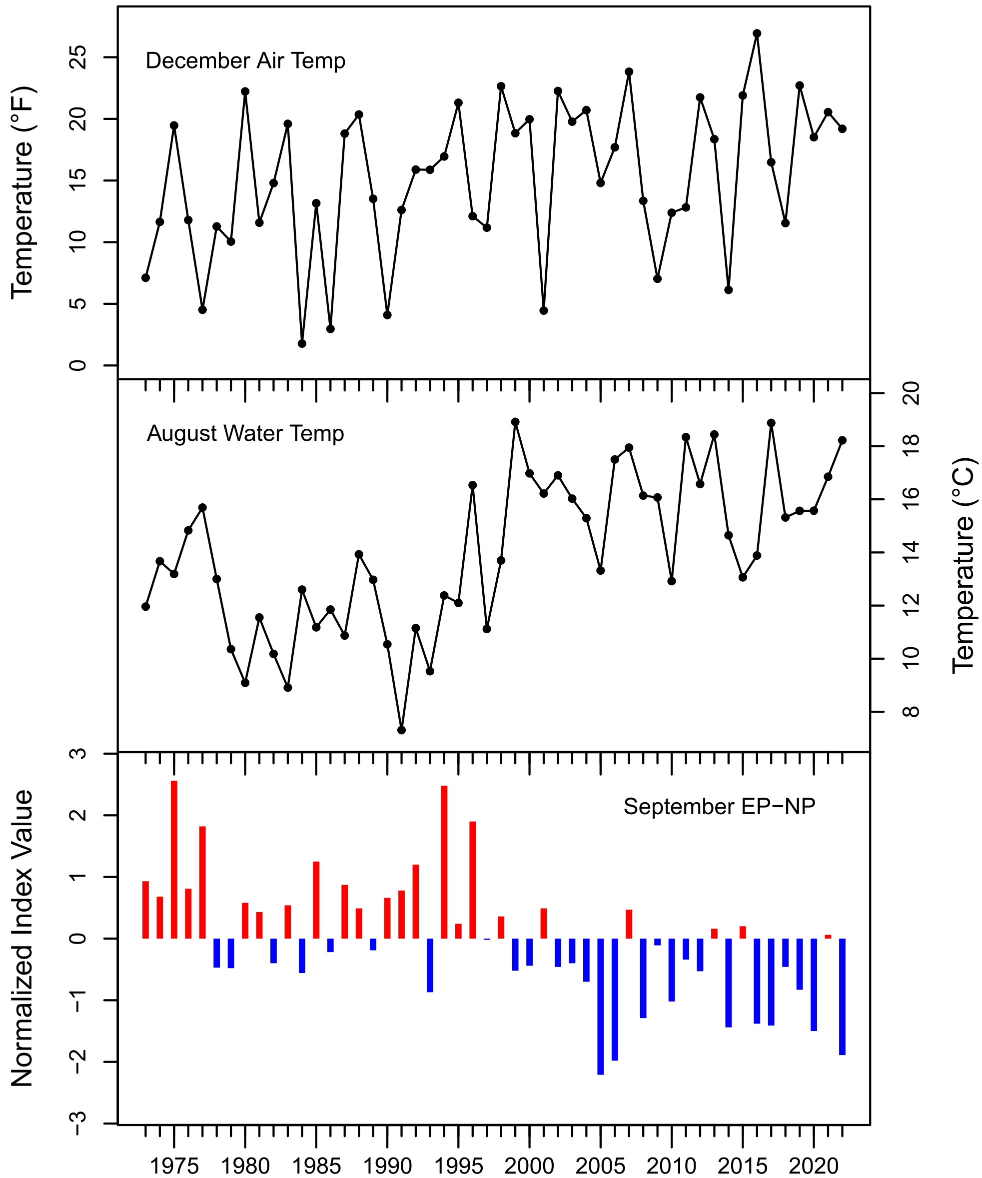

Figure 4. Time series of climatological variables that have explanatory power in our Cox (December T_a and September EP-NP) and beta models (August T_w and September EP-NP) and are associated with persistent changes in APIS ice cover that began in the late 1990s. A time series of all explanatory variables in the Cox and beta models are included in Figure S5 in Supporting Information S1. Note that *x*-axis labels refer to the year of the ice season (or "ice year") for which the variables above had explanatory power.

3.3. Model Skill Assessment

An analysis of our selected Cox and beta models indicates that the beta model (top panels Figure 5) correctly indicates a high probability (greater than 0.5) of ice cover in 38 of the 40 years in which ice cover was observed (true positive rate of 0.9), and a low probability of ice cover in 6 of the 10 years in which no ice cover was observed (true negative rate of 0.6). The Cox model (bottom panels Figure 5) yields similar results for the probability of ice cover in a given season, also correctly indicating a high probability of ice cover in 35 of the 40 years in which ice cover was observed (close to 0.9 true positive rate) and correctly indicating a low probability of ice cover in 8 of the 10 years with no observed ice cover (0.8 true negative rate). These results are promising, and suggest that both the Cox and beta models, developed using only four and (respectively) two explanatory

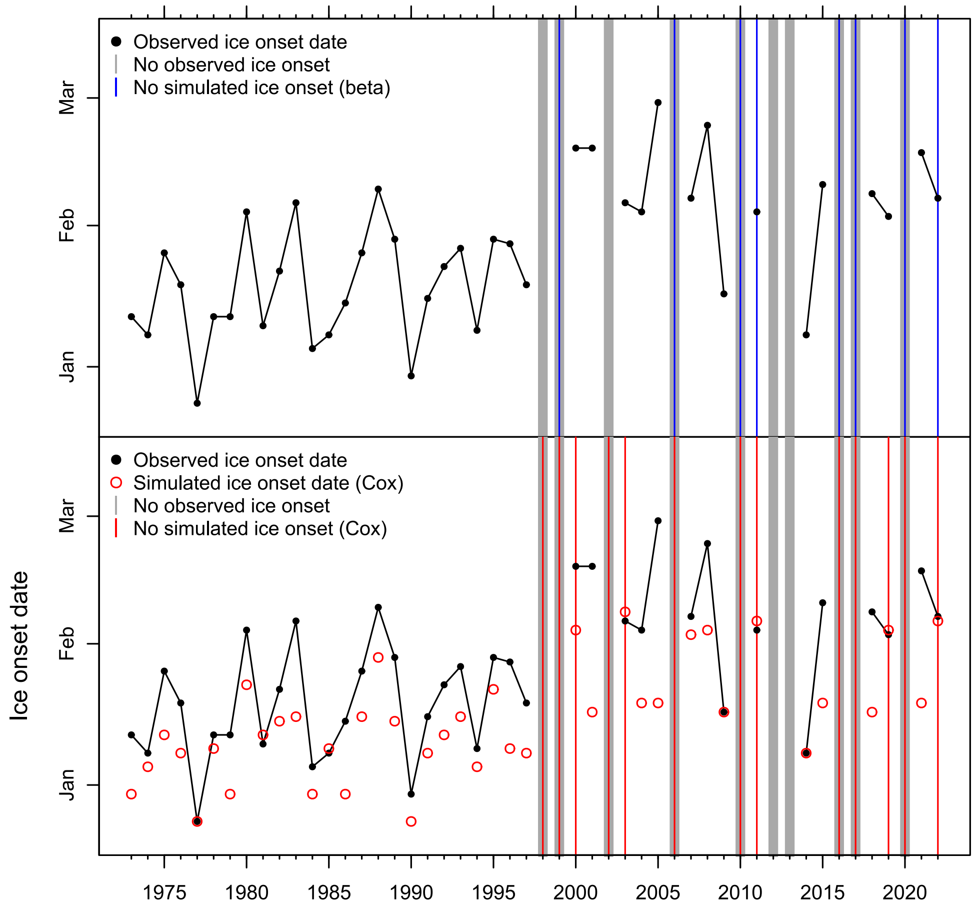

Figure 5. Results from beta (top) and Cox (bottom) simulations from final selected models (for details, see Figure S3 in Supporting Information S1) with observed ice cover onset dates (black dots). For clarity, consecutive years of observed ice cover are connected with a black line and years without observed ice cover are highlighted with a thick gray vertical line. Vertical colored lines represent years when either the beta (blue, top) or Cox (red, bottom) models indicated probability of ice cover less than 0.5. Red circles (Cox model, bottom) represent the "earliest probable ice onset" date, defined by APIS personnel as the date after which we expect there to be a 90% chance of ice onset.

variables, can provide helpful guidance to coastal management authorities about the relative probability of ice cover in a given year.

It is informative to note that in the 2 years in which the beta model incorrectly indicated a low probability of ice cover (2011 and 2022), the simulated probability of ice cover was just below 0.5 (see Figure S2 in Supporting Information S1). Similarly, of the 5 years in which the Cox model incorrectly indicated a low probability of ice cover (2000, 2003, 2011, 2019, and 2022), 2 years (2011, 2019) had simulated probabilities just below 0.5. These findings collectively suggest that a slight modification of the probability thresholds for ice cover formation could lead to improved model performance and support for management decisions.

For the years in which there was observed ice cover, the Cox survival model provided a reliable simulation of (as defined for this study) the "earliest probable onset date," by simulating an ice onset date that matched or preceded the observed ice onset date in 35 of the 40 years with observed ice cover. The five years for which the Cox model simulated ice onset after the observed date were 1981, 1985, 2003, 2011, and 2019, and in each of those years, ice cover formed very shortly before the simulated ice onset date suggesting (as with the beta model results) that a slightly different probability threshold for defining "earliest probable onset date" might lead to better model results and, importantly, more informed management actions. This modification could, of course, come with the possible trade-off that, in some years, the gap between simulated and actual ice onset might be too long, leading to overly conservative planning decisions.

The results of our cross-validation (Figure S3 in Supporting Information S1) indicate that our model is relatively robust to application across different historical periods and for potential use in forecasting. In other words, our model results were quite similar for many of the years in our study regardless of whether that year was included in model calibration, suggesting that both models are well-suited for simulating conditions in future unobserved years as well. We recognize that a component of assessing our models' suitability for future simulation and forecasting would also involve an analysis of the corresponding skill of forecasting our selected explanatory variables (Figure 3). We anticipate pursuing this additional layer of forecasting skill assessment in future research.

3.4. Climate Variables and Ice Cover Dynamics

Of the two teleconnections we found to have explanatory power for simulating ice cover (i.e., AO and EP-NP) we highlight the fact that the EP-NP pattern shows a much more (relative to AO) pronounced shift in modality in the late 1990s (Figure 5 and Figure S1 in Supporting Information S1). This shift aligns with the period when APIS ice cover appears to enter a new regime with intermittent years of no ice, later dates of ice onset, and higher variability across these and other ice cover metrics (Bai et al., 2012; Cannon et al., 2024; J. Wang, Bai, et al., 2012). Our identification of AO as a potential explanatory variable during only the pre-1998 ice cover regime (coupled with our identification of the EP-NP pattern as an explanatory variable across both regimes) may also be a manifestation of the observed increase in Lake Superior water temperatures over the course of our study (most profoundly in 1998; upper right panel Figure S5 in Supporting Information S1) and a reduction in the relative impact of changes in AO after 1998.

Further, while previous studies reflect inconsistent relationships between AO and cold air outbreaks (Huang et al., 2021; Smith & Sheridan, 2019; Yu et al., 2015), EP-NP may in fact be more significant to climate conditions (including cold air outbreaks) over the Great Lakes as fluctuating Pacific sea surface temperatures propagate through thermal amplification of the high pressure ridge over western Canada (Newton et al., 2014; Oertel et al., 2023) and across the North American continent (Stuivenvolt-Allen et al., 2023). Although the PNA pattern (which was not found to be a significant explanatory variable in our models) is also characterized by atmospheric height anomalies in western North America (Toride & Hakim, 2021), the role of the EP-NP pattern has perhaps been underestimated in regulating climate conditions in the Great Lakes region, given the coinciding changes in EP-NP, atmospheric height enhancement, and ice cover variability. This underscores its potential utility in new monthly and seasonal forecasting systems (Schulte & Lee, 2017).

From a global climate change perspective, it is informative to note that these phenomena (such as changes in teleconnections, air surface temperatures, and ice cover) are evident in the historical record from the past 50 years (IPCC, 2021) across a broad range of spatial scales. In the area surrounding Lake Superior, these broader connections to global climate change are clearly evident at the spatial scale of NOAA's "climate divisions." These divisions (Guttman & Quayle, 1996; Vose et al., 2014b) are multi-station climate zones with long representative data sets of temperature and precipitation intended to characterize a particular region within a state. In the climate division that includes the APIS (the Wisconsin Northwest climate division) maximum winter temperatures have increased by roughly 2–4° (Celsius) over the past 70 years (Figure S4 in Supporting Information S1) and, importantly, are now above freezing at a higher frequency. Average temperatures have also increased in the Wisconsin Northwest climate division over this period, though these increases are less appreciable that those of maximum temperatures.

4. Conclusions

We developed an updated set of probabilistic models for simulating and forecasting seasonal ice cover in a critical natural resource area along the southern shoreline of Earth's largest (by surface area) freshwater lake. These improved models, which build on the findings of Ji et al. (2019), incorporate an extra 7 years of ice cover data for this area and additional explanatory variables across the entire study period. Our results indicate that model improvements were critically linked to the addition of both in situ data (including antecedent air and surface water temperatures; T_a and T_w), as well as the EP-NP teleconnection pattern. Importantly, our findings indicate that the EP-NP pattern is critical to simulating the onset and persistence of ice cover near the APIS, and may be more important than previously identified teleconnections, such as AO and PDO (J. Wang et al., 2018), in explaining cold air outbreaks across the Great Lakes and other regions as well (Schulte & Lee, 2017).

We believe our study is among the first of its kind to link a pronounced shift in the EP-NP teleconnection pattern with corresponding long-term changes in both the mean and the variability of seasonal Great Lakes surface temperatures and ice cover. Other recent studies (Cannon et al., 2023) noting this potential linkage have done so with an accompanying call for additional research on the potential physical mechanisms linking the EP-NP pattern to Great Lakes climate conditions; our current study responds to that call and sets the stage for even further research on understanding the importance of teleconnection patterns linking atmospheric and oceanic conditions in the Pacific, atmospheric heights in western North America, and climate conditions (including cold air outbreaks) across both lakes (Sorensen et al., 2024) and the land surface (Shin et al., 2024) of central North America.

Our new models (based on those in Ji et al., 2019) also represent an advancement in coastal ice cover forecasting through application and testing of novel survival (Cox) and beta regression models ideally suited for representing the earliest probable date of ice cover onset and (respectively) the probability of ice cover in a given year. We know of no ice cover studies, in the Great Lakes or elsewhere, aside from Ji et al. (2019), that align the probability distribution of modeled ice-related response variables with the probability distribution of the prescribed model. Aside from the application we have provided here to improve understanding of multi-decadal coastal ice cover dynamics along the coastline of Earth's largest freshwater lake, we believe these two models have the potential to be utilized, with recalibration, in coastal areas around the world, and we view continued application of our methodology as an important priority for future research.

These findings collectively advance understanding of interannual and multi-decadal ice cover dynamics, and provide a robust basis for forecasting ice cover conditions, especially across forecast horizons longer than those currently being explored (i.e., medium-range; up to 15 days) by state-of-the-art mechanistic models (Yeo et al., 2024). Our findings are all-the-more important because they show that model capabilities were improved by isolating additional key driving factors of an apparent regime shift in Great Lakes regional ice cover and surface temperature dynamics that occurred in the late 1990s (Assel, 1998; Rodionov & Assel, 2003; Van Cleave et al., 2014). Both physics-based and statistical models often fail to represent these types of regime shifts and, therefore, ultimately fail to correctly simulate and forecast environmental conditions, because they do not fully represent changes in the key driving mechanisms (Biggs et al., 2009; Müller et al., 2014). By addressing this current gap in research, our methodology and results provide a robust basis for improving not only management planning at the APIS, but seasonal ice cover forecasting along other coastal regions as well.

Data Availability Statement

All data sets analyzed in this study are publicly available at the following sites:

Ice cover: https://www.glerl.noaa.gov/data/ice/glicd/grids/grid1973.zip.

Reference: Yang et al. (2020).

SOI: https://psl.noaa.gov/data/correlation/soi.data.

Reference: National Weather Service Climate Prediction Center (2024b).

NAO: www.esrl.noaa.gov/psd/data/correlation/nao.data.

Reference: Climate Research Unit University of East Anglia (2024).

(updated from Jones et al., 1997).

ENSO: www.esrl.noaa.gov/psd/gcos_wgsp/Timeseries/Data/nino34.long.anom.data.

Reference: NOAA Physical Sciences Laboratory (2024a) and Rayner et al. (2003).

AO: www.cpc.ncep.noaa.gov/products/precip/CWlink/daily_ao_index/

monthly.ao.index.b50.current.ascii.

Reference: National Weather Service Climate Prediction Center (2024a).

PDO: www.esrl.noaa.gov/psd/gcos_wgsp/Timeseries/Data/pdo.long.data.

Reference: NOAA Physical Sciences Laboratory (2024b).

PNA: www.esrl.noaa.gov/psd/data/correlation/pna.data.

Reference: National Weather Service Climate Prediction Center (2024d).

EP-NP: https://psl.noaa.gov/data/correlation/epo.data.

Reference: National Weather Service Climate Prediction Center (2024c).

AWSSI: https://mrcc.purdue.edu/AWSSI/RAWSSI/rawssi.DLHthr.csv.

Reference: Midwest Regional Climate Center (2024).

Lake temperature (post-1995): https://coastwatch.glerl.noaa.gov/statistics/average-surface-water-temperature-glsea/.

Reference: NOAA CoastWatch Great Lakes Regional Node (2024).

Lake temperature (pre-1995): https://deepblue.lib.umich.edu/data/anonymous_link/show/eac71801a07f97571-b8e01da4c511227d5caf4feafe0506b8437c39082d6940b?locale=en.

Reference: No additional reference information.

Air temperature: www.ncdc.noaa.gov/cdo-web/datasets/GHCND/stations/GHCND:USW00014913/detail.

Reference: NOAA National Center for Environmental Information (2024).

Climate Division temperature: https://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.ncdc:C00005.

Reference: Vose et al. (2014a).

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